

Anand International College of Engineering
Subject – Data Science with AI & MLops

Date : 05/06/2026

Topic – Basic Python & Python data types

Assignment 1:

1. Create a string variable with your name and perform the following operations:
 - o Print the first character
 - o Print the last character
 - o Print the length of the string
 - o Convert the string to uppercase
 - o Convert the string to lowercase
 - o Reverse the string using slicing
2. Write a Python program to perform string slicing on a given string:
 - o Print the first four characters
 - o Print characters from index 2 to 5
 - o Print the complete string in reverse order
3. Create a list containing numbers and perform the following operations:
 - o Add an element using append()
 - o Insert an element at a specific position
 - o Remove an element using remove()
 - o Delete the last element using pop()
 - o Reverse the list
 - o Sort the list

- o Find the length of the list
 - o Count the occurrence of an element
4. Create a tuple containing 5 subjects. Perform the following operations:
- o Print the first element
 - o Print the last element
 - o Print the length of the tuple
 - o Print elements using slicing
 - o Find the maximum, minimum, and sum of numerical tuple elements
5. Write a Python program to demonstrate tuple packing and unpacking. Create a tuple of four values and assign them to four separate variables.
6. Create a dictionary containing student details:
- o Name
 - o Age
 - o Course
 - o Address

Perform the following operations:

- o Print all keys
 - o Print all values
 - o Print all items
 - o Update the Address
 - o Add a new key called "Branch"
7. Write a Python program to access an element from a nested list.

Example:

[1, 2, 3, 4, [2, 5], 7]

Print the value 5 using indexing.

8. Write a Python program to take a number as input from the user and:
 - o Add 10 using the += operator
 - o Print the updated value
9. Write a Python program to demonstrate type casting:
 - o Take two numbers as input from the user
 - o Convert them to integers
 - o Print their multiplication
10. Create a dictionary and demonstrate the use of:
 - o `get()`
 - o `keys()`
 - o `values()`
 - o `items()`
11. Create a copy of a list using `copy()` and print both the original and copied lists.

Note: These Projects are meant strictly for your own learning and practice. Students are advised **not to copy answers from any website or online platform.**

Assignment submission form link -

<https://forms.gle/YYE3mTudHYwEwWEh8>